

Argus — Optics, Exposure & Measurement Field Guide

Lens hardware, sensor controls, and the interaction between them

IMX585 (B0498) · Canon EF 75–300 mm · Jetson Orin Nano · Rev 1.0 · 2 August 2026

WHAT THIS GUIDE IS FOR

The bench troubleshooting guide covers getting pixels out of the camera. This one covers getting *correct* pixels out of it — and, more importantly, knowing that they are correct.

The rig has two independent control layers. One lives on the lens barrel and is invisible to software. The other lives in V4L2 and is invisible to the eye. Most of the failures documented here came from changing something in one layer while attributing the result to the other. Section 5 is the heart of the document; the rest exists to make Section 5 legible.

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1 Scope and Reading Order

This guide assumes the camera enumerates at SuperSpeed and delivers frames. If it does not, start with the bench troubleshooting guide.

Three audiences, three routes through the document:

IF YOU ARE...	READ
Setting up on the roof and want numbers	§8 (DOPE book), Appendix A tables
Getting an unexpected measurement	§5 (interaction), §6 (measurement validity)
New to photographic controls	§2 → §3 → §4 in order

Every numeric table in Appendix A is derived from the sensor geometry below rather than quoted from a datasheet, so the derivations can be re-run when the optics change.

QUANTITY	VALUE	SOURCE
Pixel pitch	2.90 μm	IMX585 datasheet
Active area (3840×2160)	11.136 × 6.264 mm	pitch × pixel count
Diagonal	12.777 mm	computed
Crop factor vs full-frame	3.386×	43.267 / 12.777
CoC, viewing criterion	0.00852 mm	diagonal / 1500
CoC, 1-pixel criterion	0.00290 mm	= pitch
CoC, 2-pixel criterion	0.00580 mm	= 2 × pitch

2 Photographic Fundamentals for Machine Vision

2.1 Three controls, one quantity

Exposure is the total light collected per frame. Three controls set it, and only three:

CONTROL	LAYER	WHAT IT CHANGES	SIDE EFFECT
Aperture (f-number)	Mechanical ring	How wide the light path opens	Depth of field, diffraction, corner sharpness
Shutter (exposure time)	V4L2 <code>exposure_time_absolute</code>	How long the sensor integrates	Motion blur, frame-rate ceiling
Gain (ISO)	V4L2 <code>gain</code>	Amplification after capture	Noise

Gain is the odd one out: it does not collect more light, it multiplies what was already collected — signal and noise together. Aperture and shutter are the only controls that change how many photons actually arrive.

2.2 Stops: the universal currency

A **stop** is a factor of two in light. Every control is measured in the same currency, which is what makes them tradeable: open the aperture one stop and you can halve the exposure time for identical brightness.

STOPS	LIGHT FACTOR	F-NUMBERS	SHUTTER	GAIN VALUE
0	1×	f/1.4	1/1000 s	100
1	2×	f/2	1/500 s	200
2	4×	f/2.8	1/250 s	400
3	8×	f/4	1/125 s	800
4	16×	f/5.6	1/60 s	1600
5	32×	f/8	1/30 s	3200
6	64×	f/11	1/15 s	—
7	128×	f/16	1/8 s	—

The f-number sequence looks arbitrary but is not: each step multiplies by $\sqrt{2}$, because aperture *area* scales with the square of the diameter. The gain column assumes `gain=100` is unity — see §4.5, which is not yet confirmed.

WHY THIS MATTERS OPERATIONALLY

Going from f/16 to f/2.8 is 7 stops — a factor of 128. That single ring movement is larger than the entire useful range of the gain control (about 5.3 stops). The mechanical layer has more authority over exposure than the software layer does, which is exactly why leaving it undocumented is fatal.

2.3 What each control costs

Aperture buys light, spends depth of field and resolution

Opening up shortens depth of field, so focus becomes critical. Closing down eventually invokes diffraction, which softens the whole frame uniformly (§3.5). There is a sweet spot, usually one to two stops below wide open.

Shutter buys light, spends motion tolerance

Longer integration means anything that moves — the target, the rig, the atmosphere — smears across pixels. It also caps frame rate: exposure can never exceed one frame period (§4.3).

Gain buys brightness, spends signal-to-noise

Gain is the fallback when the other two are exhausted. It multiplies the read noise along with the signal, so small, low-contrast targets — precisely the maritime case — degrade first.

2.4 Why maritime inverts the usual priorities

In ordinary photography, shutter is the flexible control and aperture is chosen for aesthetics. On the roof, the ordering reverses:

- **Aperture is fixed by the lens.** The Canon 75–300 mm is f/5.6 at the long end. That is not a choice; it is the maximum.

- **Shutter is fixed by motion.** At ~ 2 arcsec/pixel, a vessel and any rig vibration both smear quickly. The anti-blur target is $\approx 1/250$ s.
- **Gain is therefore the only free variable**, and the noise it adds lands directly on 30–60 px targets that have no detail to spare.

This is why the gain sweep in §8.4 is not a bench curiosity. It is the low-light operating envelope of the entire system, and it belongs in the capstone's System Limitations section.

3 The Mechanical Layer — the Lens Barrel

3.1 Mount geometry and flange distance

Flange focal distance is the gap from the lens mounting surface to the sensor. It is a fixed property of the mount standard, and getting it wrong shifts the entire focus range.

MOUNT	FLANGE DISTANCE	NOTE
C-mount	17.526 mm	The stock 16 mm lens is C
CS-mount	12.526 mm	The B0498 body is CS
Difference	5.000 mm	The supplied adapter ring
Canon EF	44.00 mm	Via the Fotasy EF→C adapter

A C lens on a CS body *without* the ring sits 5 mm too close to the sensor. The focus range shifts outward past infinity, and near subjects become unreachable at any ring position. The symptom is softness that no amount of focusing corrects.

CONFIRMED ON THIS RIG

The 5 mm ring shipped pre-installed on the B0498. Flange distance is not the cause of any softness observed to date. Verify by eye before assuming otherwise — it is a 30-second check that eliminates an entire failure class.

3.2 The aperture ring

The f-number is focal length divided by entrance pupil diameter. Larger number = smaller opening = less light. The counterintuitive direction causes real errors: $f/16$ is *seven stops darker* than $f/1.4$.

On a Canon EF lens mounted through a passive adapter there is no electronic aperture control — the ring position is whatever it was physically left at, and nothing in the capture chain can read it. On the stock 16 mm C-mount lens the ring is a manual iris with click or friction detents.

DOCUMENTED FAILURE — 30 JULY BENCH SESSION

An initial capture returned luma mean 20.8/255, peak 111, and 22.4 % of pixels crushed to black. Considerable effort went into diagnosing auto-exposure behaviour and V4L2 control state. The actual cause was the aperture ring sitting at f/16 — roughly 7 stops down from the lens maximum. Re-opening to f/2.8 moved the mean to 73 with peak 253 and no clipping.

Rule: read the aperture ring and the focus ring before touching a single V4L2 control. The mechanical layer has more exposure authority than the entire software layer, and it reports nothing.

3.3 The focus ring

Focus sets the subject distance at which the image plane is sharp. Two properties matter here and both are counterintuitive at telephoto:

- **Infinity is a physical stop, not a range.** Many lenses focus slightly *past* infinity to allow for thermal expansion. Racking to the hard stop does not guarantee sharp distant focus.
- **Infinity focus cannot be validated indoors.** At 16 mm f/2.8 the hyperfocal distance is 10.8 m; at 300 mm f/5.6 it is 1.9 km. There is no room in a building to test either.

3.4 Depth of field and the circle of confusion

Depth of field is the range of distances rendered "acceptably sharp." That phrase hides a threshold — the **circle of confusion** (CoC), the largest blur spot still considered a point. Change the threshold and every DoF number changes.

Hyperfocal distance, the near limit when focused at infinity:

$$H = f^2 / (N \cdot c) + f \quad f = \text{focal length}, N = \text{f-number}, c = \text{CoC}$$

The conventional CoC is the sensor diagonal / 1500 — a criterion built around human viewing of a print. **That criterion is wrong for this project.** A detector consuming a 40 px vessel cares about per-pixel sharpness, which is a CoC of one to two pixels — roughly 3× stricter.

LENS / APERTURE	H (VIEWING)	H (2 PX)	H (1 PX)
16 mm f/1.4	21.5 m	31.5 m	63.1 m
16 mm f/2.8	10.8 m	15.8 m	31.5 m
16 mm f/16	1.9 m	2.8 m	5.5 m
75 mm f/5.6	118 m	173 m	346 m
150 mm f/5.6	472 m	693 m	1386 m
300 mm f/5.6	1887 m	2771 m	5542 m

THE OPERATIONAL CONSEQUENCE AT 300 MM

Focused at infinity with a 1-pixel criterion, nothing closer than **5.5 km** is pixel-sharp. Focused instead at the hyperfocal distance of 5.5 km, the sharp zone runs from **2.8 km to infinity** — which covers most of the shipping lane rather than only its far half.

Deliberate hyperfocal focusing is therefore a real setup decision, not a detail. Record the chosen focus mark in the DOPE book; it is not recoverable afterwards.

This also reframes the earlier bench confusion. At f/16 the hyperfocal distance is only 1.9 m, so the 3 m bookshelf was within depth of field even with the ring at "far." The f/16 frame was dark, not defocused. Opening to f/2.8 raised hyperfocal to 10.8 m and made focus genuinely matter for the first time.

3.5 Diffraction versus aberration

Two different physical limits bound sharpness, and they move in opposite directions with aperture:

- **Aberration** dominates wide open. Consumer zoom lenses are softest at maximum aperture; stopping down one to two stops improves them.
- **Diffraction** dominates stopped down. It is a hard physical limit set by the wave nature of light, and no lens quality can overcome it.

Airy disk diameter $d = 2.44 \cdot \lambda \cdot N$ ($\lambda = 550$ nm, green light)

F-NUMBER	AIRY DIAMETER	IN PIXELS (2.9 MM)	ASSESSMENT
f/1.4	1.88 μm	0.65 px	Aberration-limited
f/2.8	3.76 μm	1.30 px	Near-optimal
f/4	5.37 μm	1.85 px	Near-optimal
f/5.6	7.52 μm	2.59 px	Diffraction becoming visible
f/8	10.74 μm	3.70 px	Diffraction-limited
f/16	21.47 μm	7.40 px	Severely softened

At f/16 a point source spreads across more than seven pixels. For a 40 px vessel that is catastrophic. The Canon at f/5.6 sits at 2.6 px — acceptable, and in any case not a choice, since f/5.6 is wide open at 300 mm.

APERTURE SELECTION RULE

16 mm bench work: f/2.8 to f/4. Past the aberration floor, short of diffraction.

Canon at 300 mm: f/5.6, because that is the maximum. Never stop down for depth of field at long range — everything is already at infinity, and the resolution loss is unrecoverable.

3.6 The invisible-state problem

This is the single most important idea in the guide.

V4L2 cannot see the lens. Aperture, focus, and zoom position do not appear in `v4l2-ctl --list-ctrls`, in the environment manifest, or in the image metadata. A capture taken at f/16 and one taken at f/2.8 are byte-format identical and indistinguishable from the file alone.

Consequences worth internalising:

- A frame with no recorded aperture is **unreproducible**. Not inconvenient — unreproducible.
- Two frames whose mechanical state differed cannot be compared, even if every software control matched.
- Any measured result quoted without the barrel state is uncitable in the thesis.

The mitigation is procedural, not technical: the DOPE book (§8). Every capture row carries the mechanical state as a mandatory field. `argus_opticheck.sh` warns when it is missing.

4 The Software Layer — V4L2

4.1 Control inventory (B0498, uvcvideo)

CONTROL	RANGE	DEFAULT	NOTES
<code>auto_exposure</code>	0 / 1	0	0 = Auto, 1 = Manual. Gates the next row.
<code>exposure_time_absolute</code>	5 – 250000	5	100 µs units. Clamped by frame period.
<code>gain</code>	100 – 4000	100	Assumed ×0.01; unverified (§4.5)
<code>brightness</code>	−64 – 64	0	ISP offset, post-capture
<code>contrast</code>	0 – 20	10	ISP tone curve — leave at default
<code>saturation</code>	0 – 15	10	Irrelevant to luma metrics
<code>white_balance_automatic</code>	0 / 1	1	Set 0 for repeatable colour
<code>white_balance_temperature</code>	2300 – 6500	4500	Inactive while WB auto is on
<code>power_line_frequency</code>	0 / 1 / 2	0	Set 2 (60 Hz) for indoor work
<code>backlight_compensation</code>	0 / 1	0	STARVIS 2 low-light mode

4.2 Ordering: manual mode must come first

`exposure_time_absolute` carries the `inactive` flag while `auto_exposure` is in Auto. Writes to it are accepted at the CLI and silently discarded.

```
# WRONG – the exposure write is dropped
v4l2-ctl -d /dev/argus-cam --set-ctrl=exposure_time_absolute=400

# RIGHT – manual first, then exposure, then verify
v4l2-ctl -d /dev/argus-cam --set-ctrl=auto_exposure=1
v4l2-ctl -d /dev/argus-cam --set-ctrl=exposure_time_absolute=400
v4l2-ctl -d /dev/argus-cam --get-ctrl=auto_exposure,exposure_time_absolute,gain
```

The read-back is not optional. It is the only evidence the setting took.

4.3 The frame-period clamp

Exposure time cannot exceed one frame period — the sensor cannot integrate for longer than the interval between frames. Requests above the ceiling are silently truncated, producing a darker frame than the numbers imply.

FRAME RATE	PERIOD	MAX EXPOSURE VALUE	EQUIVALENT SHUTTER
90 fps (720p)	11.11 ms	111	1/90 s
60 fps	16.67 ms	166	1/60 s
30 fps	33.33 ms	333	1/30 s
15 fps	66.67 ms	666	1/15 s
10 fps	100.0 ms	1000	1/10 s
5 fps	200.0 ms	2000	1/5 s

SET THE FRAME RATE BEFORE THE EXPOSURE

If 1080p defaults to 60 fps, the ceiling is 166. A request of 200 or 600 is quietly clamped and the resulting frame is under-exposed for reasons invisible in the control read-back — the device may echo the requested value while the sensor uses the clamp.

```
v4l2-ctl -d /dev/argus-cam --set-parm=15      # do this first
```

4.4 Units

The UVC specification defines `exposure_time_absolute` in units of 100 μ s. Under that convention:

SHUTTER	VALUE	MILLISECONDS	USE
1/2000 s	5	0.5	Control minimum
1/1000 s	10	1.0	Bright daylight
1/500 s	20	2.0	Daylight
1/250 s	40	4.0	Anti-blur target
1/125 s	80	8.0	Overcast
1/60 s	167	16.7	Dusk — blur risk begins
1/30 s	333	33.3	Static targets only
1/15 s	667	66.7	Bench only

Arducam is not guaranteed to honour the convention. The exposure sweep in Appendix B tests it: if doubling the value does not double the light, the units are not what the specification says. *Status: unverified.*

4.5 Gain

The range 100–4000 with a default of 100 strongly suggests hundredths — 1.00× to 40.00×, about 5.3 stops. This is an inference from the range, not a documented fact. The gain sweep in Appendix B tests it directly.

4.6 Flicker

Mains-powered lighting pulses at twice the supply frequency — 120 Hz on a 60 Hz supply. When exposure time is not an integer multiple of the flicker period, frames land at different points in the cycle and brightness varies shot to shot.

```
v4l2-ctl -d /dev/argus-cam --set-ctrl=power_line_frequency=2 # 60 Hz
```

At 60 ms exposure enough cycles are averaged that flicker is invisible. At 4 ms it is not, and it will corrupt a bracketing series. Irrelevant outdoors; essential on the bench.

5 Interaction — Where the Layers Collide

5.1 The auto-exposure feedback loop

Auto-exposure is a closed loop: it measures scene brightness and adjusts shutter and gain to hit an internal target. That is helpful for viewing and **fatal for measurement**, because it makes exposure a dependent variable.

Under AE, changing anything that alters scene brightness — aperture, focus, framing, the lights — causes AE to compensate. The frame comes back near its target regardless, and the controls that produced it are unknown and unlogged.

THE RULE

No measurement taken under auto-exposure is comparable to any other measurement. Not to a previous frame, not to a later one. Set `auto_exposure=1` before the first frame of any series and do not change it mid-series.

5.2 Worked failure from the bench

The sequence below is reconstructed from the 30 July / 1 August session. It is included because every step looked reasonable at the time.

FRAME	MECHANICAL	SOFTWARE	MEAN	MAX	LAP
frame.raw	f/16, far	AE auto	20.8	111	38.9
frame8.raw	f/2.8, ~2 cm from near	AE auto	73.0	253	44.7
frame11.raw	f/2.8, ~2 cm from near	exp 200, gain 100	3.96	19	–
frame12.raw	f/2.8, ~2 cm from near	exp 600, gain 800	51.7	197	11.6

What actually happened, step by step:

1. **frame.raw** was dark because the aperture was at f/16, not because of any software fault. AE was already at its ceiling trying to compensate.
2. **frame8** opened the aperture by 7 stops (128× more light) but the mean rose only 3.5×. AE absorbed the difference by shortening the shutter or dropping gain — *by an unknown and unrecorded amount*.
3. Because AE moved between frame.raw and frame8, **the Laplacian comparison 38.9 → 44.7 measured nothing**. Two unknown exposure states.
4. **frame11** switched to manual and pinned gain to its floor of 100, discarding whatever amplification AE had been supplying. Result: near-black.
5. **frame12** recovered exposure but at 60 ms — long enough to integrate building sway and rig settling. The Laplacian fell to 11.6 and visible smearing appeared on the book spines.

THREE LESSONS, IN ORDER OF COST

1. Check the barrel first. Steps 1–2 cost an entire diagnostic cycle chasing software for a mechanical fault.
2. Lock exposure before comparing anything. Step 3 produced a number that was cited and reasoned about before anyone noticed it was meaningless.
3. Change one variable at a time. Step 4 changed aperture, focus, exposure mode, and framing between measurements.

5.3 Interaction matrix

CHANGE THIS...	...AND THIS MOVES WITH IT	MITIGATION
Aperture ring	Exposure, DoF, diffraction, corner sharpness	Lock AE first; re-bracket exposure; re-verify focus
Focus ring	Framing (focus breathing), scene contrast, AE response	Manual exposure; re-check framing between steps
Zoom ring (Canon)	Focal length, max aperture, focus point, FOV	Treat each zoom position as a separate configuration
Exposure time	Motion blur, frame-rate ceiling, flicker sensitivity	Set frame rate first; verify against the clamp table
Gain	Noise floor, Laplacian inflation	Inspect the frame, not only the metric
Frame rate	Maximum available exposure	Always set before exposure
Resolution	Pixels on target, FOV if cropped rather than binned	Resolve binning vs crop empirically
Touching the rig	Framing, focus, settling vibration	Wait 5 s after any contact

5.4 The exposure/gain trade — the key diagnostic

Holding total light constant while trading shutter against gain separates the two blur mechanisms, because motion blur depends on exposure time and noise depends on gain, and nothing else changes:

```
# Configuration A – long shutter, low gain
v4l2-ctl -d /dev/argus-cam --set-ctrl=exposure_time_absolute=600
v4l2-ctl -d /dev/argus-cam --set-ctrl=gain=800

# Configuration B – short shutter, high gain (~same brightness, 1/4 the
# motion-integration window)
v4l2-ctl -d /dev/argus-cam --set-ctrl=exposure_time_absolute=150
v4l2-ctl -d /dev/argus-cam --set-ctrl=gain=3200
```

RESULT	DIAGNOSIS
B sharper than A	Motion blur. Something is moving — target, rig, or building.
A and B equal	Focus error or optical limit. Run a focus sweep.
B noticeably grainier but edges crisper	Both: motion blur was present, and you are now near the gain ceiling.

6 Measurement — Luma and Laplacian

6.1 Exposure statistics

In YUYV 4:2:2, every even-indexed byte is a luma sample. Four numbers characterise exposure:

STATISTIC	TARGET	FAILURE MEANING
Mean	90 – 140	Low = under-exposed; high = over-exposed
Max	230 – 250	Below 200 = headroom wasted; 255 = clipping
Clipped ($Y \geq 250$)	< 1 %	Highlights unrecoverable — sun glint on water
Crushed ($Y \leq 5$)	< 5 %	Shadows unrecoverable — dark hulls lost

Mean alone is not sufficient. A scene that is genuinely dark in part — a shadowed shelf, an unlit hull — produces a low mean honestly. Max and the clip/crush pair are the more reliable indicators of whether the sensor's range is being used.

MARITIME ASYMMETRY

Clipping is worse than crushing here. A clipped sun-glint region is a solid white blob that reads to a detector as a high-contrast object — the specular-clutter false positive documented in the thesis problem statement. Expose to protect the highlights and accept a lower mean.

6.2 Laplacian variance

The Laplacian is a second-derivative operator: it responds to how fast intensity changes are changing. Sharp edges give large values, smooth gradients near zero. Taking the variance over the frame produces a single sharpness number.

$$L(x,y) = I(x+1,y) + I(x-1,y) + I(x,y+1) + I(x,y-1) - 4 \cdot I(x,y)$$

Blur is a low-pass filter — it removes exactly the high-frequency content the Laplacian responds to. So sharper images give higher variance, and the metric works as a focus peak-finder.

6.3 What it does not measure

Three failure modes, all encountered on this project:

It scales with contrast, roughly as the square

A dark frame yields a low value regardless of focus. `frame.raw` (mean 20.8) and `frame8` (mean 73.0) cannot be compared: the exposure difference alone predicts a change of nearly an order of magnitude.

It is diluted by empty scene content

Blank wall and shadow contribute near-zero. `frame8` measured 44.7 across the full frame despite legible book spines and crisp box lettering, because roughly half the frame was featureless. **Always crop to a detailed region.**

It rewards noise

Sensor noise is high-frequency content. At high gain the value rises while the image gets worse. This makes the metric actively misleading in exactly the dusk regime the project cares about.

VALID USE

Laplacian variance is **only** valid as a peak-finder within a single sweep where framing, aperture, exposure, and gain are all locked and only focus moves. It is not an absolute quality score and must never be compared across sessions.

6.4 Normalisation and cropping

Dividing by the image's own variance cancels most of the contrast dependence and makes values roughly comparable across exposure levels:

```
normalised = var(Laplacian) / var(luma)
```

Combined with a centre crop over a detailed region, this is the metric `argus_frame_stats.py` reports as `lap_norm`. Validation against synthetic targets: a sharp reference frame gives 5.35, the same frame blurred with a 9×9 kernel gives 0.155 — a 34× separation.

6.5 The gamma trap

The camera's ISP applies a display gamma of roughly 2.2 before the luma reaches YUYV. **Mean luma is therefore not proportional to light.**

Doubling the actual light does not double the mean; it multiplies it by about $2^{1/2.2} \approx 1.37$. Any attempt to verify control linearity from raw luma means will find severe apparent sublinearity that is entirely an artefact of the encoding.

```
argus_frame_stats.py frame.raw --linear    # removes gamma before statistics
```

Use `--linear` for linearity sweeps. Use the default for exposure judgement, since the clip and crush thresholds are defined in encoded space.

6.6 Valid-comparison protocol

1. Set the frame rate first (`--set-parm`).
2. Set `auto_exposure=1` . Verify by read-back.
3. Set `white_balance_automatic=0` and `power_line_frequency=2` .
4. Set exposure and gain. Verify by read-back. Check against the clamp table.
5. Record aperture, focus mark, and focal length by hand.
6. Lock the tripod. Do not touch it again for the duration of the series.
7. Wait 5 s after any physical contact before capturing.
8. Change **one** variable per step.
9. Use `--stream-count=1 --stream-skip=9` so the file is exactly one frame.
10. Measure with a centre crop over a detailed region.
11. Log every row to the DOPE book, including the failures.

CONFIRMED: SKIP AND COUNT ADD, THEY DO NOT OVERLAP

`--stream-count=10 --stream-skip=9` produces a file of exactly 10 frames, not 1. Measured: 41,472,000 B at 1920×1080, which is 10.00 frames to the byte. Use `--stream-count=1` when a single frame is wanted, or every statistic is silently averaged across ten.

7 Encode and Decode Paths

7.1 Hardware inventory — Orin Nano

BLOCK	PRESENT	USE
NVDEC (decode)	Yes	<code>nvv4l2decoder</code> , <code>h264_nvdec</code> , <code>nvjpegdec</code>
NVENC (encode)	No	Absent on Orin Nano. H.264/265 encode is CPU-bound.
NVJPG	Yes	<code>nvjpegenc</code> — the stills and MJPEG path

7.2 The capability-listing trap

NVIDIA ships one ffmpeg build across the whole Jetson family, compiled with NVENC support because Orin NX and AGX have the block. On this device:

```
$ ffmpeg -hide_banner -encoders | grep -i nv
V...D av1_nvenc      NVIDIA NVENC av1 encoder
V...D h264_nvenc     NVIDIA NVENC H.264 encoder
V...D hevc_nvenc     NVIDIA NVENC hevc encoder
V...D nvjpegenc      NVIDIA nvJPEG encoder
```

Three of those four cannot work here. Listing a codec means the binary knows how to address that hardware, **not that the hardware exists**. Calling `h264_nvenc` fails at runtime with a device-open error.

A RECURRING CLASS OF ERROR ON THIS PROJECT

This is the second instance of a listing being mistaken for a fact. `apt-cache show` reports repository availability, not installed state. `ffmpeg -encoders` reports build configuration, not silicon.

Rule: a capability listing is never a capability check. Exercise the path and observe the result before recording it.

7.3 Raw YUYV

A `.raw` capture is bare pixel bytes: no header, no dimensions, no format. Every tool touching it must be told the geometry explicitly.

RESOLUTION	BYTES PER FRAME	MIB
1280 × 720	1,843,200	1.76
1920 × 1080	4,147,200	3.96
3840 × 2160	16,588,800	15.82

File size is the fastest integrity check available: divide by the expected frame size. A non-integer result means truncation or wrong geometry. A result of 10.00 means the skip/count interaction from §6.6.

7.4 Where to run what

TASK	HOST	WHY
Capture	Argus	The camera is there
Frame extraction to PNG	Argus	Reduce before transfer — ship 10 MB, not 3.5 GB
Luma / Laplacian metrics	Argus	Enables in-field correction while still on site
Visual review (<code>ffplay</code>)	WSL2	Argus is headless; X-forwarding 4K raw is unusable
Training	SuperPOD	Per advisor guidance

Two WSL2 notes: keep working files in the ext4 home rather than `/mnt/c/`, because the 9p crossing is slow at these sizes; and the hardware-accelerated `ffmpeg` build matters only for MJPEG and H.264 work — raw YUYV involves no codec, so that workflow is identical on stock Ubuntu `ffmpeg`.

8 The DOPE Book

8.1 The analogy

A sniper's DOPE book — Data On Previous Engagements — records every shot: range, wind, temperature, ammunition lot, the adjustment dialled, and where the round actually landed. Nobody derives a firing solution from first principles under time pressure. They look up conditions that resemble today's and start from what worked.

The Argus problem is structurally identical. Optimal exposure at 300 mm f/5.6 on a hazy September afternoon with the sun at 240° is not worth re-deriving on the roof while the light changes. It is worth *looking up*.

The parallel that matters most: a DOPE book records the **conditions** alongside the **setting** alongside the **result**. Any one of the three alone is useless.

8.2 What to record

GROUP	FIELDS	SOURCE
Identity	UTC timestamp, session ID, filename	Automatic
Mechanical	Lens, focal length, f-stop, focus mark	Manual — invisible to software
Software	Resolution, fps, exposure value, gain	Automatic via <code>--get-ctrl</code>
Conditions	Sun azimuth/elevation, cloud, visibility, sea state, temp, humidity	Manual
Target	Range, vessel class, approximate size	Manual / AIS
Result	Luma mean, max, clip %, crush %, lap, lap_norm	Automatic
Notes	Anything unusual	Manual

The mechanical group is the one that will be skipped under time pressure and the one whose absence destroys the row. `argus_opticheck.sh` warns when `ARGUS_APERTURE` or `ARGUS_FOCUS` is unset.

8.3 Session workflow

Pre-session — before leaving

1. Confirm SuperSpeed enumeration and free NVMe space.
2. Set `SESSION`, `ARGUS_LENS`, `ARGUS_FOCAL`.
3. Note forecast conditions and sun azimuth for the window.
4. Review the DOPE book for prior rows under similar conditions — that is the point of keeping it.

On arrival

1. Mount, level, lock. Splay the legs wide, keep the column low, weight the hook.
2. Set the aperture ring. **Write it down.**
3. Set the frame rate, then manual exposure, WB, and flicker.
4. Run `argus_opticheck.sh status` and confirm every line.
5. Focus sweep on a fixed distant target; record the winning mark.
6. Bracket exposure against current light; record the winning value.

Per capture

1. Update `ARGUS_RANGE` and `ARGUS_COND` if they have changed.
2. Capture. Wait 5 s after any contact with the rig.
3. Check clip and crush before moving on — correctable on site, not afterwards.

Post-session

1. Commit the DOPE CSV to the capture repository alongside the frames.
2. Re-run the environment manifest if anything about the software stack changed.
3. Add a row to the conditions lookup (§8.4).

8.4 Building the lookup table

After several sessions the DOPE book supports the query it exists for: *given today's conditions, what settings worked last time?*

CONDITIONS	FOCAL / F-STOP	EXPOSURE	GAIN	NOTES
Bright sun, clear, sun behind	300 mm f/5.6	–	–	to be measured
Bright sun, glint on lane	300 mm f/5.6	–	–	protect highlights; expect clipping
Overcast	300 mm f/5.6	–	–	to be measured
Haze / low visibility	300 mm f/5.6	–	–	contrast loss, not exposure loss
Civil twilight	300 mm f/5.6	–	–	gain ceiling likely binds here
Night, lit vessels only	300 mm f/5.6	–	–	anti-blur target may be unreachable

The table is deliberately empty. Filling it is a field-testing deliverable, and the rows populate themselves from the DOPE CSV.

WHY THIS IS A CAPSTONE CONTRIBUTION, NOT HOUSEKEEPING

The bottom two rows define where the system stops working. The gain level at which small-target detail degrades sets the dusk cutoff, and the light level at which 1/250 s becomes unreachable sets the night cutoff. Both are exactly the kind of honest boundary the committee looks for in System Limitations — and both come directly out of the DOPE book rather than requiring separate experiments.

One caution for the exposure figures: any latency or throughput number quoted alongside them is void without the power mode. The Orin Nano was in 15 W mode for all bench measurements to date. Record `nvpmodel` state in the session header.

A Reference Tables

A.1 Field of view and angular resolution

FOCAL LENGTH	H FOV	V FOV	ARCSEC / PX	SCENE WIDTH @ 5 KM
16 mm (stock)	38.38°	22.15°	37.39	3480 m
75 mm (Canon wide)	8.49°	4.78°	7.98	742 m
150 mm	4.25°	2.39°	3.99	371 m
300 mm (Canon tele)	2.13°	1.20°	1.99	186 m

A.2 Pixels on target at 300 mm, 4K

RANGE	3 M SKIFF	12 M WORKBOAT	100 M COASTER	300 M CONTAINER
1 km	310	1241	10345	31034
2 km	155	621	5172	15517
5 km	62	248	2069	6207
8 km	39	155	1293	3879

Geometric best case. Real usable pixels are lower after atmospheric contrast loss, residual mount jitter (amplified 3.39× by the crop factor), and any defocus. The 3 m column crosses the thesis "feature collapse" threshold of 32 px between 5 and 8 km.

A.3 Exposure value quick reference

SHUTTER	VALUE	MS	MAX FPS	CONTEXT
1/2000 s	5	0.5	any	Control minimum
1/1000 s	10	1.0	any	Bright sun on water
1/500 s	20	2.0	any	Daylight
1/250 s	40	4.0	any	Anti-blur target
1/125 s	80	8.0	90	Overcast
1/60 s	167	16.7	59	Dusk; blur risk begins
1/30 s	333	33.3	30	Static targets only
1/15 s	667	66.7	15	Bench only

A.4 Depth of field, 16 mm bench work

APERTURE	HYPERFOCAL	DOF AT 3 M	NOTE
f/1.4	21.5 m	2.63 – 3.48 m	Aberration-limited; focus critical
f/2.8	10.8 m	2.35 – 4.16 m	Recommended bench setting

APERTURE	HYPERFOCAL	DOF AT 3 M	NOTE
f/5.6	5.4 m	1.93 – 6.76 m	Forgiving; slight diffraction
f/16	1.9 m	1.16 m – ∞	Focus irrelevant; badly diffraction-limited

Viewing criterion (CoC = diagonal/1500). At the 1-pixel criterion every hyperfocal figure roughly triples — see §3.4.

A.5 Frame-period clamp

FPS	PERIOD	MAX EXPOSURE VALUE	LONGEST SHUTTER
90	11.11 ms	111	1/90 s
60	16.67 ms	166	1/60 s
30	33.33 ms	333	1/30 s
15	66.67 ms	666	1/15 s
10	100.0 ms	1000	1/10 s
5	200.0 ms	2000	1/5 s

A.6 Open questions

QUESTION	TEST	STATUS
Does exposure follow the 100 μ s UVC convention?	Exposure sweep, <code>--linear</code>	Unverified
Is <code>gain</code> in hundredths (100 = 1.00×)?	Gain sweep, <code>--linear</code>	Inferred from range
Is 1080p binned or centre-cropped from 4K?	FOV comparison at fixed framing	Open — affects pixels on target
Is the ISP gamma actually 2.2?	Grey-step wedge	Assumed
Does MJPEG firmware exist for this unit?	Vendor query	Pending Arducam response

B Code Blocks

B.1 Locking exposure — the canonical sequence

```
DEV=/dev/argus-cam

# 1. Frame rate FIRST – it sets the exposure ceiling
v4l2-ctl -d $DEV --set-parm=15

# 2. Manual mode BEFORE exposure – the control is inactive under AE
v4l2-ctl -d $DEV --set-ctrl=auto_exposure=1

# 3. Exposure and gain
v4l2-ctl -d $DEV --set-ctrl=exposure_time_absolute=400
v4l2-ctl -d $DEV --set-ctrl=gain=800

# 4. Remove the remaining automatic variables
v4l2-ctl -d $DEV --set-ctrl=white_balance_automatic=0
v4l2-ctl -d $DEV --set-ctrl=power_line_frequency=2

# 5. VERIFY – the only evidence any of it took
v4l2-ctl -d $DEV --get-ctrl=auto_exposure,exposure_time_absolute,gain
```

B.2 Single-frame capture

```
# --stream-skip discards frames so AE/ISP can settle; --stream-count is
# how many are then WRITTEN. They add. Use count=1 for one frame.
v4l2-ctl -d /dev/argus-cam \
    --set-fmt-video=width=1920,height=1080,pixelformat=YUYV \
    --stream-mmap --stream-count=1 --stream-skip=9 \
    --stream-to=frame.raw

# Integrity check: must be exactly 4,147,200 bytes
stat -c%s frame.raw
```

B.3 Exposure statistics

```
python3 -c "
import numpy as np
y = np.fromfile('frame.raw', dtype=np.uint8)[0::2]
print(f'Luma mean:  {y.mean():.1f} / 255')
print(f'Min / max:  {y.min()} / {y.max()}')
print(f'Clipped (Y>=250): {(y>=250).mean()*100:.1f}%')
print(f'Crushed (Y<=5):  {(y<=5).mean()*100:.1f}%')
"
```

B.4 Laplacian variance with centre crop

```
python3 -c "
import numpy as np
```

```

W, H = 1920, 1080
fb = W * H * 2
raw = np.fromfile('frame.raw', dtype=np.uint8)
print(f'frames in file: {raw.size / fb:.2f}')
y = raw[:fb].reshape(H, W*2)[: , 0::2].astype(np.float32)
c = y[H//2-200:H//2+200, W//2-300:W//2+300]
lap = (c[1:-1,2:] + c[1:-1,:-2] + c[2:,1:-1] + c[:-2,1:-1] - 4*c[1:-1,1:-1])
print(f'raw variance: {lap.var():.1f}')
print(f'normalised: {lap.var()/c.var():.5f}')
"

```

B.5 Exposure bracket

```

DEV=/dev/argus-cam
v4l2-ctl -d $DEV --set-parm=15
v4l2-ctl -d $DEV --set-ctrl=auto_exposure=1
v4l2-ctl -d $DEV --set-ctrl=gain=100

for E in 20 40 80 160 320 640; do
    v4l2-ctl -d $DEV --set-ctrl=exposure_time_absolute=$E
    v4l2-ctl -d $DEV --set-fmt-video=width=1920,height=1080,pixelformat=YUYV \
        --stream-mmap --stream-count=1 --stream-skip=9 --stream-to=e_$E.raw
    python3 -c "
import numpy as np
y = np.fromfile('e_$E.raw', dtype=np.uint8)[0::2]
print(f'exp $E -> mean {y.mean():6.2f} max {y.max():3d}')
"
done

```

B.6 Gain bracket

```

DEV=/dev/argus-cam
v4l2-ctl -d $DEV --set-parm=15
v4l2-ctl -d $DEV --set-ctrl=auto_exposure=1
v4l2-ctl -d $DEV --set-ctrl=exposure_time_absolute=200

for G in 100 200 400 800 1600 3200 4000; do
    v4l2-ctl -d $DEV --set-ctrl=gain=$G
    v4l2-ctl -d $DEV --set-fmt-video=width=1920,height=1080,pixelformat=YUYV \
        --stream-mmap --stream-count=1 --stream-skip=9 --stream-to=g_$G.raw
    python3 argus_frame_stats.py g_$G.raw --csv
done

```

B.7 Motion-blur versus focus discriminator

```

DEV=/dev/argus-cam
# A: long shutter, low gain
v4l2-ctl -d $DEV --set-ctrl=exposure_time_absolute=600 --set-ctrl=gain=800
v4l2-ctl -d $DEV --set-fmt-video=width=1920,height=1080,pixelformat=YUYV \
    --stream-mmap --stream-count=1 --stream-skip=9 --stream-to=A_long.raw

# B: short shutter, high gain – comparable brightness, 1/4 the motion window
v4l2-ctl -d $DEV --set-ctrl=exposure_time_absolute=150 --set-ctrl=gain=3200
v4l2-ctl -d $DEV --set-fmt-video=width=1920,height=1080,pixelformat=YUYV \
    --stream-mmap --stream-count=1 --stream-skip=9 --stream-to=B_short.raw

```

```
python3 argus_frame_stats.py A_long.raw
python3 argus_frame_stats.py B_short.raw
# B sharper -> motion blur. Equal -> focus or optical limit.
```

B.8 Raw YUYV to PNG (reduce before transfer)

```
# On Argus – extract one frame, ship 10 MB instead of 3.5 GB
ffmpeg -f rawvideo -pixel_format yuyv422 -video_size 1920x1080 \
  -i frame.raw -frames:v 1 frame.png

# Frame N from a multi-frame burst
ffmpeg -f rawvideo -pixel_format yuyv422 -video_size 3840x2160 -framerate 15 \
  -i reference_15s.yuv -vf "select=eq(n\,30)" -frames:v 1 frame030.png

# On WSL2 – playback (Argus is headless)
ffplay -f rawvideo -pixel_format yuyv422 -video_size 1920x1080 frame.raw
```

B.9 Verifying hardware encode paths

```
# Expected to FAIL – Orin Nano has no NVENC block
ffmpeg -hide_banner -f lavfi -i testsrc=size=640x480:rate=30 \
  -frames:v 30 -pix_fmt nv12 -c:v h264_nvenc -f null - 2>&1 | tail -5

# Expected to SUCCEED – NVJPG is present
ffmpeg -hide_banner -f rawvideo -pixel_format yuyv422 -video_size 1920x1080 \
  -i frame.raw -frames:v 1 -pix_fmt nv12 -c:v nvjpege nc hw_test.jpg 2>&1 | tail -5

# Encoder pixel-format requirements
ffmpeg -hide_banner -h encoder=nvjpege nc
```

C Diagnostic Scripts

C.1 Installation

```
cd ~/projects/computer-vision/maritime-edge/capture
chmod +x argus_opticheck.sh argus_frame_stats.py

export SESSION=bench-01
export ARGUS_LENS=16mm-stock ARGUS_FOCAL=16
export ARGUS_APERTURE=2.8 ARGUS_FOCUS='2cm-from-near'
export ARGUS_RANGE=3 ARGUS_COND='indoor-night-ceiling'

./argus_opticheck.sh status
./argus_opticheck.sh set 400 800
./argus_opticheck.sh shot
./argus_opticheck.sh sweep-exposure 100
./argus_opticheck.sh sweep-gain 200
./argus_opticheck.sh sweep-focus
```

`argus_opticheck.sh` refuses to run a focus sweep under auto-exposure, validates every exposure request against the frame-period clamp, verifies control writes by read-back, and warns when the mechanical state is undeclared. Every capture appends a row to `dope_book.csv`.

C.2 `argus_frame_stats.py`

```
#!/usr/bin/env python3
"""
argus_frame_stats.py – exposure and focus metrics for raw YUYV 4:2:2 captures.

Reads a headerless .raw/.yuv file, extracts the luma plane, and reports
exposure statistics and Laplacian-variance focus metrics.

Usage:
  argus_frame_stats.py FILE [-W 1920] [-H 1080] [-f 0] [--crop 600x400]
  argus_frame_stats.py FILE --csv          # single CSV row, for the DOPE book
  argus_frame_stats.py FILE --linear       # gamma-linearise before stats

Exit codes:
  0  ok
  1  file/geometry error
  2  file truncated (byte count not a whole number of frames)
"""

import argparse
import os
import sys

import numpy as np

# -----
# Defaults
# -----
DEFAULT_W = 1920
```

```

DEFAULT_H = 1080
DEFAULT_CROP = "600x400"
GAMMA = 2.2 # display gamma assumed by the camera ISP; see guide 6.5

# Exposure judgement thresholds (guide 6.1)
MEAN_LO, MEAN_HI = 90.0, 140.0
CLIP_MAX = 1.0      # percent
CRUSH_MAX = 5.0     # percent

# -----
# Loading
# -----
def load_luma(path, width, height, frame_index=0, linear=False):
    """Return (luma_plane_float32, n_frames). YUYV 4:2:2 is 2 bytes/pixel;
    every even-indexed byte is a luma sample."""
    frame_bytes = width * height * 2
    size = os.path.getsize(path)

    if size < frame_bytes:
        raise SystemExit(
            f"ERROR: {path} is {size} B, smaller than one {width}x{height} "
            f"frame ({frame_bytes} B). Wrong geometry, or capture failed."
        )

    n_frames = size / frame_bytes
    if abs(n_frames - round(n_frames)) > 1e-9:
        print(
            f"WARNING: {size} B is {n_frames:.3f} frames at {width}x{height}. "
            f"File is truncated or the geometry is wrong.",
            file=sys.stderr,
        )

    n_whole = int(size // frame_bytes)
    if frame_index >= n_whole:
        raise SystemExit(
            f"ERROR: requested frame {frame_index}, file holds {n_whole}."
        )

    raw = np.fromfile(
        path, dtype=np.uint8, count=frame_bytes, offset=frame_index * frame_bytes
    )
    y = raw.reshape(height, width * 2)[: , 0::2].astype(np.float32)

    if linear:
        # Undo display gamma to approximate relative scene luminance.
        y = 255.0 * np.power(y / 255.0, GAMMA)

    return y, n_frames

# -----
# Metrics
# -----
def exposure_stats(y):
    return {
        "mean": float(y.mean()),
        "min": int(y.min()),
        "max": int(y.max()),
        "clipped_pct": float((y >= 250).mean() * 100.0),
        "crushed_pct": float((y <= 5).mean() * 100.0),
    }

```

```

}

def laplacian(y):
    """4-neighbour discrete Laplacian, interior pixels only."""
    return (
        y[1:-1, 2:] + y[1:-1, :-2] + y[2:, 1:-1] + y[:-2, 1:-1] - 4.0 * y[1:-1, 1:-1]
    )

def focus_stats(y, label):
    lap = laplacian(y)
    var = float(lap.var())
    scene_var = float(y.var())
    # Normalising by the image's own variance cancels most of the dependence
    # on scene contrast and exposure level (guide 6.4).
    norm = var / scene_var if scene_var > 1e-6 else 0.0
    return {
        f"{label}_lap_var": var,
        f"{label}_scene_var": scene_var,
        f"{label}_lap_norm": norm,
    }

def center_crop(y, cw, ch):
    h, w = y.shape
    cw, ch = min(cw, w), min(ch, h)
    y0 = (h - ch) // 2
    x0 = (w - cw) // 2
    return y[y0 : y0 + ch, x0 : x0 + cw]

# -----
# Reporting
# -----

def verdict(ex):
    notes = []
    if ex["mean"] < MEAN_LO:
        notes.append("UNDER: mean below target band")
    elif ex["mean"] > MEAN_HI:
        notes.append("OVER: mean above target band")
    if ex["clipped_pct"] > CLIP_MAX:
        notes.append(f"CLIPPED: {ex['clipped_pct']:.1f}% highlights lost")
    if ex["crushed_pct"] > CRUSH_MAX:
        notes.append(f"CRUSHED: {ex['crushed_pct']:.1f}% shadows lost")
    if ex["max"] < 200:
        notes.append(f"RANGE: peak only {ex['max']}/255, headroom unused")
    return notes or ["OK: within target band"]

def report(path, y_full, y_crop, n_frames, args):
    ex_f = exposure_stats(y_full)
    ex_c = exposure_stats(y_crop)
    fs_f = focus_stats(y_full, "full")
    fs_c = focus_stats(y_crop, "crop")

    print(f"=== {os.path.basename(path)}")
    print(f"    geometry    {args.width}x{args.height} YUYV 4:2:2")
    print(f"    frames      {n_frames:.2f} in file, analysing index {args.frame}")
    if args.linear:
        print(f"    linearised   gamma {GAMMA} removed")

```

```

print()
print("--- EXPOSURE (full frame)")
print(f"    luma mean    {ex_f['mean']:7.2f} / 255")
print(f"    min / max    {ex_f['min']:3d} / {ex_f['max']:3d}")
print(f"    clipped      {ex_f['clipped_pct']:6.2f} %    (Y >= 250)")
print(f"    crushed      {ex_f['crushed_pct']:6.2f} %    (Y <= 5)")
for n in verdict(ex_f):
    print(f"    >> {n}")
print()
print(f"--- FOCUS (crop {y_crop.shape[1]}x{y_crop.shape[0]} centred)")
print(f"    crop mean    {ex_c['mean']:7.2f} / 255")
print(f"    lap var      {fs_c['crop_lap_var']:9.2f}    <- compare within a sweep only")
print(f"    scene var    {fs_c['crop_scene_var']:9.2f}")
print(f"    normalised   {fs_c['crop_lap_norm']:9.5f}    <- compare across exposures")
print()
print(f"--- FOCUS (full frame, for reference)")
print(f"    lap var      {fs_f['full_lap_var']:9.2f}")
print(f"    normalised   {fs_f['full_lap_norm']:9.5f}")
print()
print("    NOTE: Laplacian variance is scene-dependent and rewards sensor")
print("          noise. Valid only as a peak-finder within one locked-down")
print("          sweep where framing, aperture and exposure are unchanged.")

def csv_row(y_full, y_crop):
    ex = exposure_stats(y_full)
    fs = focus_stats(y_crop, "crop")
    fields = [
        f"{ex['mean']:.2f}",
        f"{ex['max']}",
        f"{ex['clipped_pct']:.2f}",
        f"{ex['crushed_pct']:.2f}",
        f"{fs['crop_lap_var']:.2f}",
        f"{fs['crop_lap_norm']:.5f}",
    ]
    print(",".join(fields))

# -----
def main():
    p = argparse.ArgumentParser(description=__doc__,
                                formatter_class=argparse.RawDescriptionHelpFormatter)
    p.add_argument("file")
    p.add_argument("--width", type=int, default=DEFAULT_W)
    p.add_argument("--height", type=int, default=DEFAULT_H)
    p.add_argument("-f", "--frame", type=int, default=0,
                    help="zero-based frame index within a multi-frame file")
    p.add_argument("--crop", default=DEFAULT_CROP,
                    help="centre crop WxH for the focus metric (default 600x400)")
    p.add_argument("--csv", action="store_true",
                    help="emit one CSV row: mean,max,clip,crush,lap,lapnorm")
    p.add_argument("--linear", action="store_true",
                    help="remove display gamma before computing statistics")
    args = p.parse_args()

    try:
        cw, ch = (int(v) for v in args.crop.lower().split("x"))
    except ValueError:
        raise SystemExit(f"ERROR: --crop wants WxH, got '{args.crop}'")

    y, n_frames = load_luma(args.file, args.width, args.height,

```

```

        args.frame, args.linear)
    y_crop = center_crop(y, cw, ch)

    if args.csv:
        csv_row(y, y_crop)
    else:
        report(args.file, y, y_crop, n_frames, args)

if __name__ == "__main__":
    main()

```

C.3 argus_opticheck.sh

```

#!/usr/bin/env bash
# argus_opticheck.sh – optics + exposure diagnostics for the Argus rig.
#
# Companion to argus_camcheck.sh (link/enumeration) and argus_frame_stats.py
# (measurement). This script owns the interaction between the mechanical
# controls on the lens barrel and the V4L2 controls on the sensor.
#
# status          report link, format, controls, and clamp headroom
# set             apply exposure/gain in the correct order, verify readback
# shot            capture one frame, measure it, append a DOPE row
# sweep-exposure  bracket exposure at fixed gain
# sweep-gain      bracket gain at fixed exposure
# sweep-focus     interactive focus-ring sweep at locked exposure
#
# The mechanical state (aperture, focus mark, focal length) is INVISIBLE to
# V4L2. It must be supplied by hand or the DOPE log is worthless. See guide 3.6.

set -uo pipefail

# -----
# Configuration
# -----
VID_PID="04b4:0498"
DEV="${ARGUS_DEV:-}"
WIDTH="${ARGUS_W:-1920}"
HEIGHT="${ARGUS_H:-1080}"
FPS="${ARGUS_FPS:-15}"
SKIP="${ARGUS_SKIP:-9}"
CROP="${ARGUS_CROP:-600x400}"
OUTDIR="${ARGUS_OUTDIR:-./captures}"
DOPE="${ARGUS_DOPE:-./dope_book.csv}"
STATS="${ARGUS_STATS:-$(dirname "$0")/argus_frame_stats.py}"

# Mechanical state – no way to read these back from the device.
LENS="${ARGUS_LENS:-16mm-stock}"
FOCAL="${ARGUS_FOCAL:-16}"
APERTURE="${ARGUS_APERTURE:-unset}"
FOCUS_MARK="${ARGUS_FOCUS:-unset}"
RANGE_M="${ARGUS_RANGE:-unset}"
CONDITIONS="${ARGUS_COND:-bench}"
NOTES="${ARGUS_NOTES:-}"

C_OK=$'\033[0;32m'; C_WARN=$'\033[0;33m'; C_ERR=$'\033[0;31m'
C_HDR=$'\033[1;36m'; C_OFF=$'\033[0m'
[[ -t 1 ]] || { C_OK=""; C_WARN=""; C_ERR=""; C_HDR=""; C_OFF=""; }

```

```

ok() { printf ' %s[ OK ]%s %s\n' "$C_OK" "$C_OFF" "$*"; }
warn() { printf ' %s[WARN]%s %s\n' "$C_WARN" "$C_OFF" "$*"; }
err() { printf ' %s[FAIL]%s %s\n' "$C_ERR" "$C_OFF" "$*"; }
hdr() { printf '\n%s=== %s%s\n' "$C_HDR" "$*" "$C_OFF"; }
note() { printf ' %s\n' "$*"; }

# -----
# Device resolution
# -----
resolve_device() {
    [[ -n "$DEV" ]] && { [[ -e "$DEV" ]] || { err "ARGUS_DEV=$DEV does not exist"; exit 1; };
return; }
    [[ -e /dev/argus-cam ]] && { DEV=/dev/argus-cam; return; }

    local node
    for node in /dev/video*; do
        [[ -e "$node" ]] || continue
        local info vid pid
        info=$(udevadm info --query=property --name="$node" 2>/dev/null) || continue
        vid=$(sed -n 's/^ID_VENDOR_ID=//p' <<<"$info")
        pid=$(sed -n 's/^ID_MODEL_ID=//p' <<<"$info")
        [[ "${vid}:${pid}" == "$VID_PID" ]] || continue
        # Skip metadata-only nodes: keep the first with a capture format.
        v4l2-ctl -d "$node" --list-formats 2>/dev/null | grep -q "'YUYV'" || continue
        DEV="$node"; return
    done
    err "no device matching $VID_PID with a YUYV capture format"
    exit 1
}

# -----
# Helpers
# -----
get_ctrl() { v4l2-ctl -d "$DEV" --get-ctrl="$1" 2>/dev/null | awk -F': ' '{print $2}' | tr -d ' '; }

# Exposure is in 100 us units, and cannot exceed one frame period.
max_exposure_for_fps() { awk -v f="$1" 'BEGIN{printf "%d", int(10000/f)}'; }
exposure_ms() { awk -v e="$1" 'BEGIN{printf "%.2f", e/10}'; }
exposure_frac() { awk -v e="$1" 'BEGIN{if(e>0) printf "1/%d", 10000/e; else printf "n/a"}'; }
gain_x() { awk -v g="$1" 'BEGIN{printf "%.2f", g/100}'; }

link_speed() {
    local sysdev
    for sysdev in /sys/bus/usb/devices/*; do
        [[ -f "$sysdev/idVendor" && -f "$sysdev/idProduct" ]] || continue
        local v p
        v=$(<"$sysdev/idVendor"); p=$(<"$sysdev/idProduct")
        [[ "${v}:${p}" == "$VID_PID" ]] || continue
        [[ -f "$sysdev/speed" ]] && { cat "$sysdev/speed"; return; }
    done
    echo "unknown"
}

require_mechanical() {
    local missing=0
    [[ "$APERTURE" == unset ]] && { warn "aperture not declared – export ARGUS_APERTURE=2.8";
missing=1; }
    [[ "$FOCUS_MARK" == unset ]] && { warn "focus mark not declared – export ARGUS_FOCUS='2cm-from-near'"; missing=1; }
    [[ $missing -eq 1 ]] && note "Recording without these makes the row unreproducible."
}

```

```

    return 0
}

# -----
# status
# -----
cmd_status() {
    hdr "LINK"
    local sp; sp=$(link_speed)
    case "$sp" in
        5000|10000) ok "USB SuperSpeed: ${sp}M" ;;
        480)      err "USB 2 fallback (480M) – replug the camera; 4K will not enumerate" ;;
        *)        warn "link speed: $sp" ;;
    esac
    ok "device: $DEV"

    hdr "FORMAT"
    local cur_fps
    cur_fps=$(v4l2-ctl -d "$DEV" --get-parm 2>/dev/null | sed -n 's/.*Frames per second: \
([0-9.]*\).*\/\1/p')
    v4l2-ctl -d "$DEV" --get-fmt-video 2>/dev/null | sed -n 's/^\t/ /p' | head -5
    note "frame rate: ${cur_fps:-unknown} fps"

    hdr "EXPOSURE CLAMP HEADROOM"
    note "Exposure cannot exceed one frame period, whatever value you write."
    printf '          %-8s %-14s %-12s %s\n' "fps" "period" "max value" "= shutter"
    local f
    for f in 15 30 60 90; do
        local mx; mx=$(max_exposure_for_fps "$f")
        printf '          %-8s %-14s %-12s %s\n' \
            "$f" "$(awk -v x="$f" 'BEGIN{printf "%.2f ms", 1000/x}')" "$mx" "$(exposure_frac "$mx")"
    done
    local mx_now; mx_now=$(max_exposure_for_fps "${cur_fps:-15}")
    note ""
    note "At ${cur_fps:-15} fps your ceiling is exposure_time_absolute=$mx_now."

    hdr "SENSOR CONTROLS"
    local ae exp gain wb plf
    ae=$(get_ctrl auto_exposure); exp=$(get_ctrl exposure_time_absolute)
    gain=$(get_ctrl gain); wb=$(get_ctrl white_balance_automatic)
    plf=$(get_ctrl power_line_frequency)

    if [[ "$ae" == "1" ]]; then
        ok "auto_exposure = 1 (Manual) – measurements are valid"
    else
        err "auto_exposure = ${ae:-?} (Auto) – AE will move exposure between shots"
        note "Any A/B comparison taken like this is uninterpretable. See guide 5.1."
    fi
    ok "exposure_time_absolute = ${exp:-?} (${exposure_ms "${exp:-0}") ms, ${exposure_frac "${exp:-0}") s)"
    [[ -n "$exp" && "$exp" -gt "$mx_now" ]] && \
        err "requested $exp exceeds the ${cur_fps:-15} fps ceiling of $mx_now – silently clamped"
    ok "gain = ${gain:-?} (${gain_x "${gain:-100}")x)"
    [[ "$wb" == "1" ]] && warn "white_balance_automatic = 1 – colour will drift between shots"
    [[ "$plf" == "0" ]] && warn "power_line_frequency = 0 (Disabled) – set 2 for 60 Hz indoors"

    hdr "MECHANICAL STATE (not readable from the device)"
    note "lens          $LENS"
    note "focal length   ${FOCAL} mm"
    note "aperture        f/${APERTURE}"
    note "focus mark      $FOCUS_MARK"

```

```

    note ""
    note "V4L2 cannot see the barrel. If these are wrong, every number below"
    note "is attached to the wrong configuration. See guide 3.6."
}

# -----
# set
# -----
cmd_set() {
    local want_exp="$1" want_gain="$2"
    hdr "APPLYING"

    v4l2-ctl -d "$DEV" --set-parm="$FPS" >/dev/null 2>&1
    ok "frame rate -> $FPS fps"

    local mx; mx=$(max_exposure_for_fps "$FPS")
    if (( want_exp > mx )); then
        err "exposure $want_exp exceeds the $FPS fps ceiling of $mx"
        note "Lower the frame rate or the exposure. Clamping silently otherwise."
        return 1
    fi

    # ORDER MATTERS: exposure_time_absolute is flagged inactive under AE.
    v4l2-ctl -d "$DEV" --set-ctrl=auto_exposure=1 >/dev/null 2>&1
    ok "auto_exposure -> 1 (Manual) [must precede exposure write]"

    v4l2-ctl -d "$DEV" --set-ctrl=exposure_time_absolute="$want_exp" >/dev/null 2>&1
    v4l2-ctl -d "$DEV" --set-ctrl=gain="$want_gain" >/dev/null 2>&1
    v4l2-ctl -d "$DEV" --set-ctrl=white_balance_automatic=0 >/dev/null 2>&1
    v4l2-ctl -d "$DEV" --set-ctrl=power_line_frequency=2 >/dev/null 2>&1

    hdr "READBACK"
    local got_exp got_gain
    got_exp=$(get_ctrl exposure_time_absolute); got_gain=$(get_ctrl gain)

    if [[ "$got_exp" == "$want_exp" ]]; then
        ok "exposure_time_absolute = $got_exp ($(exposure_ms "$got_exp") ms, $(exposure_frac
"$got_exp") s)"
    else
        err "exposure: wrote $want_exp, device reports $got_exp – clamped or rejected"
    fi
    if [[ "$got_gain" == "$want_gain" ]]; then
        ok "gain = $got_gain ($(gain_x "$got_gain")x)"
    else
        err "gain: wrote $want_gain, device reports $got_gain"
    fi
}

# -----
# capture
# -----
capture_raw() {
    local dest="$1"
    v4l2-ctl -d "$DEV" \
        --set-fmt-video=width="$WIDTH",height="$HEIGHT",pixelformat=YUYV \
        --stream-mmap --stream-count=1 --stream-skip="$SKIP" \
        --stream-to="$dest" >/dev/null 2>&1
    local want=$(( WIDTH * HEIGHT * 2 ))
    local got; got=$(stat -c%s "$dest" 2>/dev/null || echo 0)
    if [[ "$got" != "$want" ]]; then
        err "capture wrote $got B, expected $want B"
    fi
}

```

```

        return 1
    fi
    return 0
}

dope_header() {
    [[ -f "$DOPE" ]] && return
    echo
    "utc,session,lens,focal_mm,f_stop,focus_mark,range_m,conditions,res,fps,exp_val,exp_ms,gain_val,gain_x,luma_mean,luma_max,clip_pct,crush_pct,lap_var,lap_norm,file,notes" > "$DOPE"
}

dope_append() {
    local file="$1" extra_notes="${2:-}"
    local exp gain metrics
    exp=$(get_ctrl exposure_time_absolute); gain=$(get_ctrl gain)
    metrics=$(python3 "$STATS" "$file" -W "$WIDTH" -H "$HEIGHT" --crop "$CROP" --csv 2>/dev/null)
    [[ -z "$metrics" ]] && { err "measurement failed for $file"; return 1; }

    dope_header
    # metrics is already the comma-joined block: mean,max,clip,crush,lap,lapnorm
    printf '%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s\n' \
        "$(date -u +%Y-%m-%dT%H:%M:%SZ)" "${SESSION:-adhoc}" \
        "$LENS" "$FOCAL" "$APERTURE" "$FOCUS_MARK" "$RANGE_M" "$CONDITIONS" \
        "$WIDTH" "$HEIGHT" "$FPS" \
        "$exp" "$(exposure_ms $exp)" "$gain" "$(gain_x $gain)" \
        "$metrics" "$(basename "$file")" "${NOTES}${extra_notes}" >> "$DOPE"
}

cmd_shot() {
    require_mechanical
    mkdir -p "$OUTDIR"
    local ae; ae=$(get_ctrl auto_exposure)
    [[ "$ae" != "1" ]] && warn "auto_exposure is not Manual – this row will not be reproducible"

    local stamp dest
    stamp=$(date -u +%Y%m%dT%H%M%SZ)
    dest="$OUTDIR/shot_${stamp}.raw"

    hdr "CAPTURE"
    capture_raw "$dest" || return 1
    ok "wrote $dest"

    hdr "MEASUREMENT"
    python3 "$STATS" "$dest" -W "$WIDTH" -H "$HEIGHT" --crop "$CROP"

    dope_append "$dest" && ok "DOPE row appended to $DOPE"
}

# -----
# sweeps
# -----
sweep_report_header() {
    printf '\n          %-8s %-10s %-9s %-8s %-8s %-10s %s\n' \
        "$1" "mean" "max" "clip%" "crush%" "lap" "lap_norm"
    printf '          %s\n' "-----"
}

sweep_row() {
    local label="$1" file="$2"
    local m; m=$(python3 "$STATS" "$file" -W "$WIDTH" -H "$HEIGHT" --crop "$CROP" --csv 2>/dev/null)

```

```

IFS=, read -r mean mx clip crush lap lapn <<<"$m"
printf '          %-8s %-10s %-9s %-8s %-8s %-10s %s\n' \
        "$label" "$mean" "$mx" "$clip" "$crush" "$lap" "$lapn"
}

cmd_sweep_exposure() {
    require_mechanical
    mkdir -p "$OUTDIR"
    local gain="${1:-100}"
    local mx; mx=$(max_exposure_for_fps "$FPS")
    hdr "EXPOSURE SWEEP (gain fixed at $gain = $(gain_x "$gain")x, $FPS fps, ceiling $mx)"
    note "Doubling the value should double the light. Where it stops doubling,"
    note "you have found the clamp, the noise floor, or a nonlinear response."
    sweep_report_header "exp"

    local e
    for e in 20 40 80 160 320 640; do
        (( e > mx )) && { printf '          %-8s CLAMPED at %s fps (ceiling %s)\n' "$e" "$FPS"
"$mx"; continue; }
        cmd_set "$e" "$gain" >/dev/null 2>&1
        local dest="$OUTDIR/sweep_exp_${e}.raw"
        capture_raw "$dest" || continue
        sweep_row "$e" "$dest"
        NOTES="sweep-exposure" dope_append "$dest" >/dev/null 2>&1
    done
    note ""
    note "Target: mean 90-140, clip < 1%, crush < 5%, max near 250 but not at it."
}

cmd_sweep_gain() {
    require_mechanical
    mkdir -p "$OUTDIR"
    local exp="${1:-200}"
    hdr "GAIN SWEEP (exposure fixed at $exp = $(exposure_ms "$exp") ms)"
    note "This sweep IS the low-light budget. On the roof, shutter is pinned"
    note "short by vessel motion, so gain is the only variable left. Where the"
    note "image degrades here is where dusk operation ends. See guide 8.4."
    sweep_report_header "gain"

    local g
    for g in 100 200 400 800 1600 3200 4000; do
        cmd_set "$exp" "$g" >/dev/null 2>&1
        local dest="$OUTDIR/sweep_gain_${g}.raw"
        capture_raw "$dest" || continue
        sweep_row "$g" "$dest"
        NOTES="sweep-gain" dope_append "$dest" >/dev/null 2>&1
    done
    note ""
    note "CAUTION: sensor noise inflates the Laplacian. A rising lap number at"
    note "high gain is noise, not sharpness. Look at the frame, not just the row."
}

cmd_sweep_focus() {
    require_mechanical
    mkdir -p "$OUTDIR"
    local ae; ae=$(get_ctrl auto_exposure)
    if [[ "$ae" != "1" ]]; then
        err "auto_exposure is not Manual. A focus sweep under AE is invalid:"
        note "racking focus changes scene contrast, AE reacts, exposure moves,"
        note "and you measure focus and AE response tangled together."
        return 1
    fi
}

```

```

fi
hdr "FOCUS SWEEP (exposure locked, framing must not move)"
note "Do not touch the tripod. After each ring adjustment, wait 5 s for the"
note "rig to settle before pressing Enter, or you log settling vibration."
sweep_report_header "position"

local i=1
while true; do
    read -r -p "          focus mark (blank to finish): " mark
    [[ -z "$mark" ]] && break
    sleep 5
    local dest="$OUTDIR/sweep_focus_$(printf '%02d' "$i")_${mark//\//-}.raw"
    capture_raw "$dest" || continue
    sweep_row "$mark" "$dest"
    FOCUS_MARK="$mark" NOTES="sweep-focus" dope_append "$dest" >/dev/null 2>&1
    i=$(( i + 1 ))
done
note ""
note "Best focus is the peak of lap_norm. If the curve is flat, the softness"
note "is optical or motion, not focus error."
}

# -----
usage() {
    sed -n '2,16p' "$0" | sed 's/^# \{0,1\} //'
    cat <<'EOF'

Environment:
    ARGUS_DEV, ARGUS_W, ARGUS_H, ARGUS_FPS, ARGUS_SKIP, ARGUS_CROP
    ARGUS_OUTDIR, ARGUS_DOPE, ARGUS_STATS
    ARGUS_LENS, ARGUS_FOCAL, ARGUS_APERTURE, ARGUS_FOCUS, ARGUS_RANGE
    ARGUS_COND, ARGUS_NOTES, SESSION

Examples:
    export ARGUS_APERTURE=2.8 ARGUS_FOCUS='2cm-from-near' SESSION=bench-01
    ./argus_opticheck.sh status
    ./argus_opticheck.sh set 400 800
    ./argus_opticheck.sh shot
    ./argus_opticheck.sh sweep-exposure 100
    ./argus_opticheck.sh sweep-gain 200
    ./argus_opticheck.sh sweep-focus
EOF
}

main() {
    command -v v4l2-ctl >/dev/null || { err "v4l2-ctl not found (apt install v4l-utils)"; exit 1; }
    [[ -f "$STATS" ]] || { err "measurement engine not found at $STATS"; exit 1; }

    local cmd="${1:-status}"; shift || true
    case "$cmd" in
        status)          resolve_device; cmd_status ;;
        set)             resolve_device; cmd_set "${1:?exposure value}" "${2:-100}" ;;
        shot)            resolve_device; cmd_shot ;;
        sweep-exposure)  resolve_device; cmd_sweep_exposure "${1:-100}" ;;
        sweep-gain)      resolve_device; cmd_sweep_gain "${1:-200}" ;;
        sweep-focus)     resolve_device; cmd_sweep_focus ;;
        -h|--help|help)  usage ;;
        *)               err "unknown command: $cmd"; usage; exit 1 ;;
    esac
    echo
}

```

main "\$@"